

Math 465: Introduction to Combinatorics

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Homework #2 is posted and due on Tuesday, September 15.

These slides will be posted on Canvas.

Generating functions

Definition

Let

$$h_0, h_1, h_2, h_3, \dots$$

be a finite (resp., infinite) sequence of nonnegative integers.

The **generating function** of this sequence is the polynomial (resp., **formal power series**)

$$h(x) = h_0 + h_1x + h_2x^2 + h_3x^3 + \dots = \sum_k h_k x^k.$$

Example

Let $h_k = k + 1$, i.e., we look at the sequence $1, 2, 3, \dots$. Then

$$h(x) = 1 + 2x + 3x^2 + 4x^3 + \dots = \sum_k (k + 1)x^k = (1 - x)^{-2}.$$

Why generating functions?

In a typical application, h_k is an answer to an enumerative question that depends on k .

Example

(Here h_k = number of k -element subsets of an n -element set.)
The generating function for the n th row of Pascal's triangle is

$$\sum_k \binom{n}{k} x^k = (1 + x)^n.$$

The generating function is often simpler than its coefficients.

One can use algebraic manipulations with generating functions to obtain results about sequences.

Generating functions: Example 1

Problem

How many different poker hands can be dealt from a double deck of 2×52 cards? (Each poker hand consists of 5 cards.)

The two instances of each card are indistinguishable from each other.

Solution #1

Each of the 52 types of cards will be drawn either 0 or 1 or 2 times. Each potential drawing can be encoded by a 52-letter word in the alphabet $\{0, 1, 2\}$.

For a drawing to contain 5 cards, the number of occurrences of 0, 1, 2 must be either $(47, 5, 0)$ or $(48, 3, 1)$ or $(49, 1, 2)$. Hence the answer is

$$\binom{52}{47 \ 5 \ 0} + \binom{52}{48 \ 3 \ 1} + \binom{52}{49 \ 1 \ 2} = 3748160.$$

Generating functions: Example 1, continued

Problem

How many different poker hands can be dealt from a double deck of 2×52 cards?

We will solve a more general problem, for hands consisting of k cards.

Solution #2 (using generating functions)

Set $h_k = \#\{k\text{-card hands that can be dealt from a double deck}\}$.

Then

$$\begin{aligned}\sum_k h_k x^k &= (1 + x + x^2)^{52} \\ &= \underbrace{(1 + x + x^2)(1 + x + x^2) \cdots (1 + x + x^2)}_{52 \text{ factors}} \quad [\text{why?}].\end{aligned}$$

In other words, h_k is simply the coefficient of x^k in $(1 + x + x^2)^{52}$.

Generating functions: Example 2

Problem

How many ways are there to change a dollar into coins?

Solution

Let h_k denote the number of solutions of

$$n_1 + 5n_2 + 10n_3 + 25n_4 + 50n_5 + 100n_6 = k$$

in nonnegative integers. We would like to compute h_{100} . We have

$$\begin{aligned}\sum_k h_k x^k &= (1+x+x^2+\cdots)(1+x^5+x^{10}+\cdots)(1+x^{10}+x^{20}+\cdots)\cdots \\ &= \frac{1}{(1-x)(1-x^5)(1-x^{10})(1-x^{25})(1-x^{50})(1-x^{100})}\end{aligned}$$

The coefficient of x^{100} in the power series expansion of this function can be computed using any symbolic computation software.

symbolic computation software

FROM THE MAKERS OF WOLFRAM LANGUAGE AND MATHEMATICA



SeriesCoefficient[1/((1-x)(1-x^5)(1-x^10)(1-x^25)(1-x^50)(1-x^100)), {x,0,100}]

 Natural Language  Math Input 

   

Input:

series coefficient	$\frac{1}{(1-x)(1-x^5)(1-x^{10})(1-x^{25})(1-x^{50})(1-x^{100})}$	point	$x = 0$
		order	100

Result:

293

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A screenshot from wolframalpha.com

Generating functions: Example 3

The formula below was derived from Newton's Binomial Theorem:

$$(*) \quad (1-x)^{-n} = \sum_{k=0}^{\infty} \binom{n+k-1}{n-1} x^k.$$

Alternative proof of (*)

We have

$$(1-x)^{-n} = \underbrace{(1+x+x^2+\cdots) \cdots (1+x+x^2+\cdots)}_n = \sum_k \boxed{?} x^k$$

where

$$\begin{aligned} \boxed{?} &= \text{number of nonnegative integer solutions} \\ &\quad \text{of the equation } a_1 + \cdots + a_n = k. \\ &= \text{number of weak compositions of } k \text{ with } n \text{ parts} \\ &= \binom{n+k-1}{n-1}. \end{aligned}$$

Weight functions

Let $\alpha : A \rightarrow \mathbb{Z}_{\geq 0}$ be a “weight function” on a set A .

For each value k , let

$$h_k = \text{number of elements } a \in A \text{ with } \alpha(a) = k.$$

(We assume that $h_k < \infty$.)

Observation

$$h(x) \stackrel{\text{def}}{=} \sum_k h_k x^k = \sum_{a \in A} x^{\alpha(a)}.$$

We will frequently deal with generating functions defined in this way.

If A is a finite set, then the generating function $h(x)$ can be viewed as a “refinement” of $|A|$, the cardinality of A . Indeed, $h(1) = |A|$.

Multiplication Principle (abstract form)

Suppose that we have a bijection of the form

$$\begin{aligned} A &\leftrightarrow B \times C \times D \times \dots \\ a &\leftrightarrow (b, c, d, \dots). \end{aligned}$$

Multiplication Principle

$$|A| = |B| \cdot |C| \cdot |D| \dots$$

We will extend the Multiplication Principle to generating functions.

Additivity for weight functions

Suppose that we have a bijection of the form

$$\begin{aligned} A &\leftrightarrow B \times C \times D \times \cdots \\ a &\leftrightarrow (b, c, d, \dots). \end{aligned}$$

and that

$$\alpha : A \rightarrow \mathbb{Z}_{\geq 0}$$

$$\beta : B \rightarrow \mathbb{Z}_{\geq 0}$$

$$\gamma : C \rightarrow \mathbb{Z}_{\geq 0}$$

$$\delta : D \rightarrow \mathbb{Z}_{\geq 0}$$

.....

are weight functions as above satisfying the **additivity condition**

$$\alpha(a) = \beta(b) + \gamma(c) + \delta(d) + \cdots .$$

Multiplication Principle for generating functions

Multiplication Principle for generating functions is a straightforward refinement of the ordinary Multiplication Principle:

Theorem

Let

$$A \leftrightarrow B \times C \times D \times \dots$$

$$a \leftrightarrow (b, c, d, \dots)$$

$$\alpha(a) = \beta(b) + \gamma(c) + \delta(d) + \dots$$

be as above. Then

$$\sum_{a \in A} x^{\alpha(a)} = \left(\sum_{b \in B} x^{\beta(b)} \right) \left(\sum_{c \in C} x^{\gamma(c)} \right) \left(\sum_{d \in D} x^{\delta(d)} \right) \dots$$

Setting $x = 1$ recovers the ordinary multiplication principle.

Multiplication Principle for generating functions: examples

Changing money into coins (again)

Let h_k = number of ways to change k cents into coins. We set

$A = \{\text{distinct collections of several coins}\};$

$B = \{\text{collections of several pennies}\};$

$C = \{\text{collections of several nickels}\};$

and so on. Each weight function $\alpha, \beta, \gamma, \dots$ assigns to a collection of coins its value in cents. The additivity condition holds. Therefore

$$\begin{aligned}\sum_k h_k x^k &= (1+x+x^2+\dots)(1+x^5+x^{10}+\dots)(1+x^{10}+x^{20}+\dots)\dots \\ &= (1-x)^{-1}(1-x^5)^{-1}(1-x^{10})^{-1}\dots(1-x^{100})^{-1}.\end{aligned}$$

Exercise: Counting poker hands (again)

What are the sets $A, B, C, \dots$ and the weight functions $\alpha, \beta, \gamma, \dots$ in that example?